



QuESO Official Manual

Sarah Riley

May 06, 2026

CONTENTS:

1	Installation	3
2	API Reference	5
2.1	writer.py	5
2.2	aux.py	5
2.3	base.py	5
2.4	approach.py	6
2.5	logg.py	7
2.6	aia.py	7
2.7	style.py	7
2.8	multiline.py	7
2.9	products.py	7
2.10	loader.py	7
2.11	base.py	8
2.12	base.py	8

Warning

This documentation is under active development.

INSTALLATION

You can install the latest version of this package from Codeberg by

```
` pip install git+https://codeberg.com/SharkSorceress/QuES0.git `
```

Once we have a proper version, we plan to submit our package to PyPi as queso-cluster.

API REFERENCE

2.1 writer.py

exportFITS(*spectralDataset*, *labelLine*, *fname*)

2.2 aux.py

_gen_dataID(*Input*)

pick_jth_label(*labelLst*, *j*)

density_2channel(*x*, *y*, *dy*, *xsize*, *top*, *bottom*)

density_hist2d(*data*, *dy*, *top*, *bottom*)

close_factors(*number*)

almost_factors(*number*)

common_elements(*ar1*, *ar2*, *ar3*)

2.3 base.py

normZ(*dataSquare*)

normMaximum(*dataSquare*)

normContinuum(*dataSquare*, *continuumIndx*)

concatSpectra(*dataSquareLst*)

numba_histogram(*a*, *bins*, *lim*)

rotateArray(*image*, *turns*)

get_bin_edges(*bins*, *lim*)

compute_bin(*x*, *bin_edges*)

np_gradient(*f*)

minimize(*data*, *decisions*, *size*)

maximize(*data, decisions, size*)
np_all_axis0(*x*)
np_all_axis1(*x*)
similarityMetric(*x, y, type='dist', ref=0*)
curvature(*y*)
startMax(*data, k, decisions*)
startPlusPlus(*data, k, decisions*)
_calcMoment(*waveAxis, ii, jj, lineCore, dataCube, order, ref, counter=0*)
_calcFeatureDensity(*data, converge, zindx, func1*)
_calcOptimization(*k, data, decision, threshold*)
_calcElbowEntry(*data, labels*)
_calcVarianceScore(*data*)
_criteriaInertiaScore(*score*)
_calcInertiaScore(*dataSquare, labelLine*)
_criteriaSilhouetteScore(*score*)
_calcSilhouetteScore(*dataSquare, labelLine*)
_calcSingleSilhouetteScore(*data, labels, lab*)
_calcCHindex(*data, labels*)
labelGluer(*labels*)
labelReorder(*labels*)
_calcQuiescentFrame(*spectralData, spectralParams, contIndxs, progress=None*)
_calcDynamicFrame(*spectralData, dynamicScanNum, progress=None, delta=0*)

2.4 approach.py

__init__(*self, config, catalogName, instrumentObj*)
__getattr__(*self, name*)
cluster(*self, prepSquare, maskLine*)
__init__(*self, catalogBase*)
clustering(*self, prepCube, tlst, groups, intrinsicSquare=None*)

Detail

low temporal resolution clustering

2.5 logg.py

loggTimer(*func*)

wrapper(*args, **kwargs)

logg(tag, val=None, _time=None, _log=None)

duration_string(dur)

2.6 aia.py

delayAIA(fname, epochDev)

2.7 style.py

cbar_bounds(bounds)

_genColorPallet(n)

rainbow_cmap(nrange, discrete=False, nan=False)

cmap_discretize(cmap, N)

__init__(self, spaceInfo, deltas)

_mapGen(self, fig, pos, arr, flareContour=None, timeAxis=None, **kwargsDict)

2.8 multiline.py

2.9 products.py

__init__(self, epochObj, optLabels)

figure03(self)

figure04_template(self)

spectralEntry(self, ax, indx, color, wavelambda, extent, scores=None)

2.10 loader.py

__init__(self, data, home, fig)

_loadEventConfig(self, eventRunnerFname, args)

__init__(self, dataPath)

vispLoad(self, stokes=0)

irisLoad(self)

```

fissLoad(self, labels)
__init__(self, config1, config2)
visp2visp(self)
fiss2fiss(self)
__init__(self, dataPath, labels)
__init__(self, dataPath, stokes=0)
__init__(self, inputLst, runIndx)
loadSource(self, eventInput)
__init__(self, srcInput)
__init__(self, runnerInput)
__init__(self, fname, eventIndx, runIndx)
_load(self, fname)

```

2.11 base.py

```

_mainIntrinsic(config, prepSquare, lineIndx, intrinsicSkip=False)
_mainOptimization(prepSquare, labelLine, kLst=None, stageMax=2)

```

2.12 base.py

```

_runOptimalKSearch(dataSquare, funcLst, checkLst)
runPrep(dataSquare, norm='continuum', keepI0=None, maskLine=None, quSquare=None,
        **kwargs)
_runIntrinsic(nbins, data, edgeOverride=None)
runStart(k, data, start='max')
_runOptimization(k, sub_data, converge)
_findOptimalK(dataSquare, funcLst, criteriaLst, converge=1e-6)
_runLabelSort(dataSquare, labelLine)

```